

Please enter what you learned from today's lecture in at least 50 characters.

I learned the key differences between supervised and unsupervised learning, specifically, how unsupervised learning doesn't use target labels. I got a really good grasp of how to use k-means clustering to group similar data points, and how to use methods like the elbow method and silhouette analysis to determine the optimal number of clusters. I also learned about dimensionality reduction using Principal Component Analysis, which helps simplify complex datasets with tons of features into something much easier to visualize and interpret.

Please provide specific details about what you particularly liked about today's lecture.

I really enjoyed the hands-on Python notebook examples, especially applying k-means to the financial marketing dataset to see how we could realistically segment customers by age and occupation. The visualizations were also great, plotting the breast cancer dataset using just two principal components to clearly separate the malignant and benign cases, making the power of PCA click for me instantly.

If there were any parts that were unclear or areas for improvement, please provide specific details.

The math behind PCA, specifically how the covariance matrix is used to calculate eigenvalues and eigenvectors to find the principal components, was just a little bit tricky to follow. It might be helpful to spend a few extra minutes breaking down the visual intuition behind that math for those of us who are slightly rusty on linear algebra.

Regarding the instructor, please provide specific details about what you liked or disliked.

Shu Morikuni was a great instructor! He gave a really clear opening and I appreciated his detailed self-introduction about his background in robotics and AI. He was also incredibly helpful and courteous during the Q&A session at the end, specifically when answering the student question about why we'd prefer unsupervised learning over simple threshold-based rules for complex anomaly detection tasks.

Please write down any lectures or fields you would like to attend in the future.

I'm really looking forward to the next session on time series data that the instructor mentioned at the very end of class! I'd also love to dive deeper into soft clustering models, such as the Gaussian Mixture models briefly mentioned.

Please feel free to write any other comments or opinions.

Thank you for the great session! The administration reminders at the beginning of class about the resubmission portal for the competition prediction code and the Session 10 homework were super helpful, so I don't miss the deadlines.